Mathematics - K-12 Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Which ratio CANNOT form a proportion with 8 over 18?
   (a) 46 over 108
   (b) 12 over 27
   (c) 60 over 135
   (d) 4 over 9

2. Three points are chosen randomly and independently on a circle. What is the probability that all three pairwise distances between the points are less than the radius of the circle?
   (a) $\frac{1}{20}$
   (b) $\frac{1}{32}$
   (c) $\frac{1}{12}$
   (d) $\frac{1}{16}$

3. We roll a fair 6-sided die 5 times. What is the probability that we get a 6 in at most 2 of the rolls?
   (a) $\frac{125}{648}$
   (b) $\frac{25}{648}$
   (c) $\frac{625}{648}$
   (d) $\frac{1}{648}$

4. The first number in a number pattern is 28. The pattern rule is to add 14 to get the next number in the pattern. If the pattern continues, which statement is true?
   (a) All the numbers in the pattern can be divided equally by 10.
   (b) All the numbers in the pattern can be divided equally by 4.
   (c) All the numbers in the pattern can be divided equally by 8.
   (d) All the numbers in the pattern can be divided equally by 7.

5. What is the sum of all integer solutions to $|n| < |n - 3| < 9$?
   (a) -9
   (b) -14
   (c) 14
   (d) 9

True / False (10 marks)

Answer True or False

6. True or False: The number 47,520 rounds to 47,000 when rounded to the nearest thousand.

7. True or False: A worker producing parts at a constant rate can make 110 parts in 35 hours if she produces 22 parts in 7 hours.

8. True or False: The maximum value of the function f(x) = xe^{-x} is e.

9. True or False: The volume of the solid with a square cross-section and a base bounded by x^2 = 4y and y = 2 is 20 cubic units.

10. True or False: The number of heads in three tosses of a coin is not a suitable scenario for a binomial model.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. In how many ways can 9 people sit around a round table? (Two seatings are considered the same if one is a rotation of the other.)

12. If there is a 70% chance of rain on a given day, explain how you would determine the probability of it not raining, and analyze the implications of weather probabilities in daily decision-making.

End of Paper